Quantum numerical linear algebra

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Overview

- Introduction to quantum computing
- Basic linear algebra operations
- Matrix functions
- Application: Hamiltonian simulation
- Application: Linear systems of equations
- Summary

Section 1

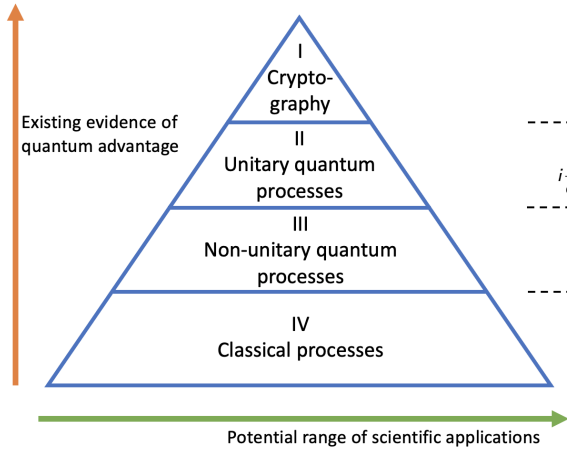
Introduction to quantum computing

Quantum computing

"If you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy." [Feynman (1982)]

- Using the nature of quantum mechanics to perform computing
- Quantum advantage:
  - Unstructured search: Quantum cost $O(\sqrt{N})$ vs. Classical cost $O(N)$
  - Fourier transform: Quantum cost $O((\log N)^2)$ vs. Classical cost $O(N \log N)$

Quantum algorithm hierarchy



Shor's algorithm (for prime factorization)

Hamiltonian simulation

$$i\frac{d}{dt}u(t) = H(t)u(t), \quad H(t) \text{ is Hermitian}$$

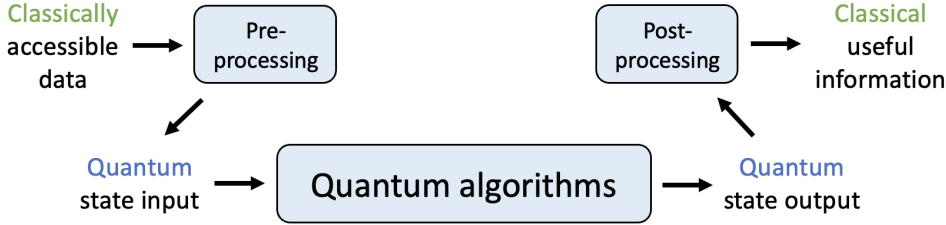
Open quantum dynamics, ground state/energy, Gibbs state preparation, Green's function

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Linear systems of equations, ODE, PDE, SDE, optimization, optimal control, machine learning

[0] Adapted from [Lin Lin. Lecture Notes on Quantum Algorithms for Scientific Computation]

Quantum algorithm flow map



Input: quantum states

A qubit (quantum bit): $\mathbb{C}^2$

- Instead of 0 or 1, quantum allows superposition $\alpha\,|0\rangle + \beta\,|1\rangle$
- Measurement: get 0 with prob $|\alpha|^2$, get 1 with prob $|\beta|^2$
- Orthonormal basis $|0\rangle = (1,0)^T$, $|1\rangle = (0,1)^T$

Multi-qubit system: $\mathbb{C}^{2^n}$, $N = 2^n$

- Superposition of 2^n possible states $|j\rangle$, $0 \le j \le 2^n - 1$:

$$|u\rangle = \sum_{j=0}^{2^n-1} \alpha_j\,|j\rangle \cong (\alpha_0, \alpha_1, \ldots, \alpha_{2^n-1})^T$$

- Tensor product: $|u\rangle = |u_A\rangle \otimes |u_B\rangle = |u_A\rangle\,|u_B\rangle$

Input: quantum states

Dirac notations:

- $|u\rangle = (\alpha_0, \ldots, \alpha_{N-1})^T \in \mathbb{C}^N$, $\| |u\rangle \| = 1$
- Computational basis: $\{|j\rangle\}_{j=0}^{N-1}$, $\quad |u\rangle = \sum_{j=0}^{N-1} \alpha_j |j\rangle$
- $\langle u| = |u\rangle^\dagger$
- Inner product: $\langle u|v\rangle$
- Observable: $\langle u| O |u\rangle$

Quantum algorithms

Quantum algorithms: Implement a series of unitary transformations

$$U_K \cdots U_2 U_1 \left|\psi_{\text{in}}\right\rangle = \left|\psi_{\text{out}}\right\rangle$$

- A quantum version of numerical linear algebra

Post-processing

$$|u\rangle = \sum_{j=0}^{N-1} \alpha_j |j\rangle$$

Measurement: collapse to $|j\rangle$ with probability $|\alpha_j|^2$

- Extract amplitude info or more general $\langle u| O |u\rangle$
- Post-select on "good subspace": $|\psi\rangle = \sqrt{p}\,|0\rangle\,|\psi_{\text{out}}\rangle + \sqrt{1-p}\,|\text{bad}\rangle$; expected repeats $O(1/p)$
- **Amplitude amplification (AA):** apply reflections, then measure; expected repeats $O(1/\sqrt{p})$

Complexity and quantum circuits

Quantum complexity:

- Query complexity: number of oracle calls
- Circuit depth: number of coherent layers

Quantum circuits: graphical representation

$$
\begin{aligned}
|1\rangle &\xrightarrow{e^{i\tilde{\phi}_d Z}} \cdots \xrightarrow{e^{i\tilde{\phi}_0 Z}} \\
|0\rangle &\xrightarrow{U_A} \xrightarrow{U_A^\dagger} \cdots \xrightarrow{U_A^{(\dagger)}} \\
&|u\rangle \cdots
\end{aligned}
$$

Section 2

Basic linear algebra operations

Input model: vectors

State preparation oracle:

$$O_u : |0\rangle \to |u\rangle = \sum_{j=0}^{2^n-1} u_j |j\rangle$$

Constructing O_u is generally hard, but efficient in special cases.[1]

[1] [arXiv:quant-ph/0208112], [arXiv:2201.11495]

Input model: matrices

Definition (Block-encoding)

Let A be $2^n \times 2^n$. A **block-encoding** of A is a $2^{n+a} \times 2^{n+a}$ unitary U_A such that

$$A \approx \alpha\big(\langle 0|^{\otimes a} \otimes I\big) U_A \big(|0\rangle^{\otimes a} \otimes I\big), \quad \text{i.e.,} \quad U_A \approx \begin{pmatrix} \frac{1}{\alpha}A & * \\ * & * \end{pmatrix}.$$

- $\alpha \geq \|A\|$ is the block-encoding factor
- Serves as both input and output model

Input model: matrices

$$U_A \approx \begin{pmatrix} \frac{1}{\alpha}A & * \\ * & * \end{pmatrix}$$

- Constructing U_A for arbitrary A is in general hard
- Special cases: unitary, sparse, structured matrices, ...[2]

[2] [arXiv:1806.01838], [arXiv:2203.10236]

Matrix-vector multiplication

Input:

Block-encoding of A:

$$A \approx \alpha \left(\langle 0| \otimes I \right) U_A \left(|0\rangle \otimes I \right)$$

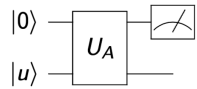
$$U_A \approx \begin{array}{c} \begin{array}{cc} \langle 0| & \langle 1| \end{array} \\ \begin{pmatrix} \frac{1}{\alpha} A & * \\ * & * \end{pmatrix} \end{array} \begin{array}{c} |0\rangle \\ |1\rangle \end{array}$$

or $$U_A \approx |0\rangle \langle 0| \otimes \frac{A}{\alpha} + |0\rangle \langle 1| \otimes * + |1\rangle \langle 0| \otimes * + |1\rangle \langle 1| \otimes *$$

Quantum state:

$$|u\rangle = \sum_{j=0}^{2^n - 1} u_j |j\rangle$$

'Algorithm': applying block-encoding



or $$U_A |0\rangle |u\rangle \approx \frac{1}{\alpha} |0\rangle A |u\rangle + c |1\rangle |*\rangle$$

or $$\begin{pmatrix} \frac{1}{\alpha} A & * \\ * & * \end{pmatrix} \begin{pmatrix} u \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\alpha} Au \\ * \end{pmatrix}$$

Need to measure the first ancilla qubit

Success probability: $(\| A |u\rangle \| / \alpha)^2$

Number of repeats (after amplitude amplification): $\mathcal{O}\left(\alpha / \| A |u\rangle \|\right)$

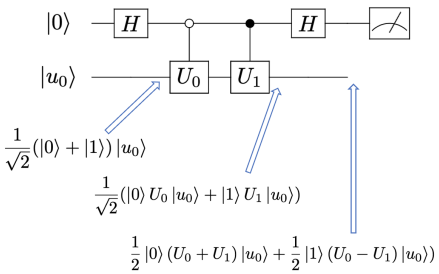
Matrix addition: Linear combination of unitaries (LCU)

Task: given unitaries $\{U_j\}$ and $c_j \in \mathbb{R}_+$, compute $\sum_j c_j U_j$

- General matrix addition: U_j's are block-encodings
- Vector addition: U_j's are state preparation oracles

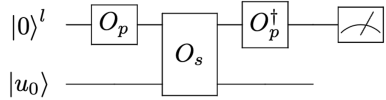
Matrix addition: LCU

A toy example: computing $\frac{1}{2}(U_0 + U_1)|u_0\rangle$



$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \begin{aligned} |0\rangle &\to \tfrac{1}{\sqrt{2}}|0\rangle + \tfrac{1}{\sqrt{2}}|1\rangle \\ |1\rangle &\to \tfrac{1}{\sqrt{2}}|0\rangle - \tfrac{1}{\sqrt{2}}|1\rangle \end{aligned}$$

General: computing $\sum_j c_j U_j$



Prepare Oracle $O_p : |0\rangle \to \frac{1}{\sqrt{\|c\|_1}} \sum_j \sqrt{c_j}\,|j\rangle$

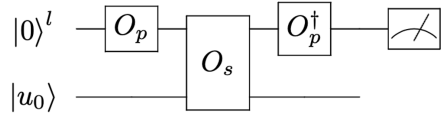
Select Oracle $O_s = \sum_j |j\rangle\langle j| \otimes U_j$

Matrix addition: LCU

$$
\begin{aligned}
|0\rangle\,|u_0\rangle &\xrightarrow{O_p} \frac{1}{\sqrt{\|c\|_1}}\sum_j \sqrt{c_j}\,|j\rangle\,|u_0\rangle \\
&\xrightarrow{O_s} \frac{1}{\sqrt{\|c\|_1}}\sum_j \sqrt{c_j}\,|j\rangle\,U_j\,|u_0\rangle \\
&\xrightarrow{O_p^\dagger} \frac{1}{\|c\|_1}\,|0\rangle\sum_j c_j U_j\,|u_0\rangle + |\perp\rangle
\end{aligned}
$$

- Repeats: $O\Big(\|c\|_1/\|\sum_j c_j U_j\,|u_0\rangle\,\|\Big)$
- Cost depends on constructing O_p, O_s

General: computing $\sum_j c_j U_j$



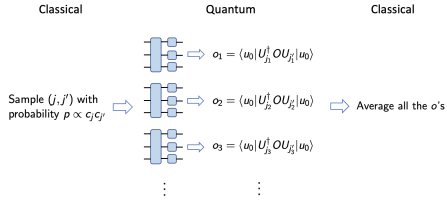
Prepare Oracle $O_p : |0\rangle \to \frac{1}{\sqrt{\|c\|_1}}\sum_j \sqrt{c_j}\,|j\rangle$

Select Oracle $O_s = \sum_j |j\rangle\langle j| \otimes U_j$

Hybrid implementation of LCU: Importance sampling

$$u = \sum_j c_j U_j \left| u_0 \right\rangle$$

$$\Rightarrow u^\dagger O u = \sum_{j,j'} c_j c_{j'} \left\langle u_0 \right| U_j^\dagger O U_{j'} \left| u_0 \right\rangle$$



Summary of basic linear algebra operations

Input models: Vector: quantum state Matrix: block-encoding

Algorithms:

- Matrix-vector multiplication
- Matrix/vector addition: LCU

Other algorithms (not covered today):

- Inner product/observable: SWAP test, Hadamard test
- Matrix multiplication

Reference: [Section 4, arXiv:1806.01838]

Section 3

Matrix functions

Matrix functions

- For Hermitian matrices: eigenvalue transformation

$$A = V\mathrm{diag}(\lambda_j)V^\dagger \longrightarrow f(A) = V\mathrm{diag}(f(\lambda_j))V^\dagger$$

- For general matrices: singular value transformation

$$A = W\mathrm{diag}(\sigma_j)V^\dagger \longrightarrow f(A) = W\mathrm{diag}(f(\sigma_j))V^\dagger$$

or $\longrightarrow V\mathrm{diag}(f(\sigma_j))V^\dagger$ or $\longrightarrow W\mathrm{diag}(f(\sigma_j))W^\dagger$

Main result

Suppose U_A block-encodes Hermitian A with $\|A\| \leq 1$, and $p(x)$ is a real polynomial s.t.

1. $\deg(p) = d$,
2. $|p(x)| \leq 1$ for all $x \in [-1, 1]$.

Then $p(A)$ can be block-encoded with complexity $O(d)$.[3]

[3] Gilyén et al. [arXiv:1806.01838]

Applications

- Linear systems: $f(x) = 1/(\kappa x)$
- Hamiltonian simulation: $f(x) = e^{-ixt}$
- Filtering
- Amplitude amplification: $U|0\rangle|\psi\rangle = \frac{1}{q}|0\rangle A|\psi\rangle + |\perp\rangle$
  $\rightarrow \tilde{U}|0\rangle|\psi\rangle = \frac{1}{2}|0\rangle A|\psi\rangle + |\perp\rangle, \quad f(x) = qx/2, \deg(p) \sim q\log(1/\epsilon)$
- ...

Toy example: Chebyshev polynomials

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x), \quad T_k(\cos\theta) = \cos(k\theta)$$

Toy example: Chebyshev polynomials

Consider 2×2 matrices (let $\theta = \arccos \lambda$):

$$U(\lambda) = \begin{pmatrix} \lambda & \sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & -\lambda \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad O = U(\lambda)Z = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Then $O^k = \begin{pmatrix} \cos(k\theta) & * \\ & * \end{pmatrix} = \begin{pmatrix} T_k(\lambda) & * \\ & * \end{pmatrix}$.

For Hermitian A: $T_k(A) = VT_k(\Lambda)V^\dagger$ — perform O^k on each 2D eigensubspace.

Qubitization

Suppose $A = \sum_j \lambda_j |v_j\rangle \langle v_j|$ and $U_A = U_A^\dagger$ is its Hermitian block-encoding.

$$U_A |0\rangle |v_j\rangle = \lambda_j |0\rangle |v_j\rangle + \sqrt{1-\lambda_j^2}\, |\perp_j\rangle, \qquad \Pi |\perp_j\rangle = 0, \quad \Pi = |0\rangle \langle 0| \otimes I.$$

Invariant space $\mathcal{H}_j = \mathrm{span}\{|0\rangle |v_j\rangle, |\perp_j\rangle\}$:

$$[U_A]_{\mathcal{H}_j} = \begin{pmatrix} \lambda_j & \sqrt{1-\lambda_j^2} \\ \sqrt{1-\lambda_j^2} & -\lambda_j \end{pmatrix}, \qquad [\Pi]_{\mathcal{H}_j} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

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[4] [arXiv:1610.06546]

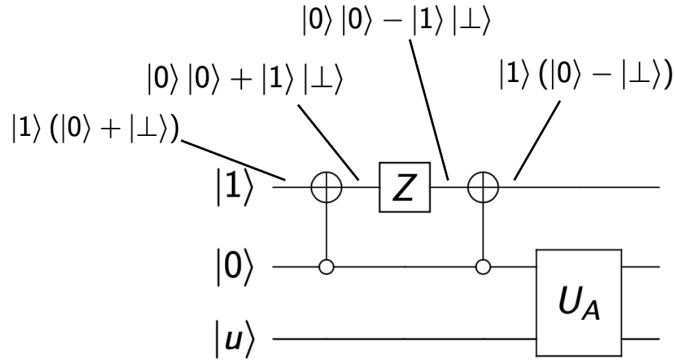
Qubitization

Let $Z_\Pi = 2\Pi - I$. Then $O = U_A Z_\Pi$ satisfies

$$[O]_{\mathcal{H}_j} = \begin{pmatrix} \lambda_j & -\sqrt{1-\lambda_j^2} \\ \sqrt{1-\lambda_j^2} & \lambda_j \end{pmatrix},$$

$$[O^k]_{\mathcal{H}_j} = \begin{pmatrix} T_k(\lambda_j) & * \\ & * \end{pmatrix}, \qquad O^k = \begin{pmatrix} T_k(A) & * \\ & * \end{pmatrix}.$$

Qubitization



$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$U_A = \begin{pmatrix} A & * \\ * & \end{pmatrix},\ U_A = U_A^\dagger,\ \Pi = |0\rangle\langle 0| \otimes I,\ Z_\Pi = 2\Pi - I$$

$$(U_A Z_\Pi)^k = \begin{pmatrix} T_k(A) & * \\ * & \end{pmatrix}$$

Qubitization

If U_A is not Hermitian (but A is): alternate between two 2D subspaces by applying U_A and $U_A^\dagger$ alternately.

Circuit: $|1\rangle \xrightarrow{Z} |0\rangle \xrightarrow{U_A} \xrightarrow{U_A^\dagger} \cdots$

Quantum signal processing (QSP)

Starting from 2×2 matrices $U(\lambda)$, Z, $O(\lambda) = U(\lambda)Z$, consider

$$U_\Phi(\lambda) = e^{i\phi_0 Z} \prod_{j=1}^{d} \left[O(\lambda) e^{i\phi_j Z} \right] = \begin{pmatrix} P(\lambda) & -Q(\lambda)\sqrt{1-\lambda^2} \\ Q^*(\lambda)\sqrt{1-\lambda^2} & P^*(\lambda) \end{pmatrix}$$

where $\Phi = (\phi_0, \phi_1, \ldots, \phi_d) \in \mathbb{R}^{d+1}$ are *phase factors*.

Quantum signal processing (QSP)

Theorem (QSP)

Phase factors $\Phi \in \mathbb{R}^{d+1}$ exist such that $U_\Phi(\lambda) = \begin{pmatrix} P & * \\ * & \end{pmatrix}$ if and only if P, Q satisfy:

1. $\deg P \leq d$, $\deg Q \leq d-1$,
2. P has parity $d \bmod 2$; Q has parity $(d-1) \bmod 2$,
3. $|P(\lambda)|^2 + (1-\lambda^2)|Q(\lambda)|^2 = 1$ for all $\lambda \in [-1, 1]$.

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[5] [arXiv:1606.02685]

QSP + Qubitization

The high-dimensional case follows via qubitization. Circuit:

$$
\begin{aligned}
|1\rangle &\xrightarrow{e^{i\tilde{\phi}_d Z}} \cdots \xrightarrow{e^{i\tilde{\phi}_0 Z}} \\
|0\rangle &\xrightarrow{U_A} \xrightarrow{U_A^\dagger} \cdots \xrightarrow{U_A^{(\dagger)}} \\
|u\rangle &\cdots
\end{aligned}
$$

Output: block-encoding of $P(A)$. Query complexity: d.

QSP for real polynomials

Theorem (QSP for real polynomials)

Given a real polynomial $p(\lambda)$ with $\deg p = d$, $d \bmod 2$ parity, $|p(\lambda)| \leq 1$ on $[-1, 1]$, there exist P, Q and phase factors $(\phi_0, \ldots, \phi_d)$ such that $\mathrm{Re}(P(\lambda)) = p(\lambda)$ and

$$U_\Phi(\lambda) = \begin{pmatrix} P(\lambda) & * \\ & * \end{pmatrix}.$$

QSP, qubitization, and QSVT

For any Hermitian A with $\|A\| \leq 1$ and block-encoding U_A, and any real polynomial p of degree d with $|p| \leq 1$ on $[-1, 1]$:

$$p(A) \text{ can be block-encoded with } O(d) \text{ queries.}$$

If A is not Hermitian, this becomes **singular value transformation (QSVT)**.[6]

[6] Gilyén et al. [arXiv:1806.01838]

Phase factors

Finding phase factors was hard when QSP was proposed; now practically solved.

- **Direct methods:** Remez exchange, roots of polynomials [arXiv:1806.01838], Capitalization [arXiv:2003.02831], Prony's method [arXiv:2202.02671]
- **Iterative methods:** optimization [arXiv:2002.11649], fixed-point iteration [arXiv:2209.10162], nonlinear Fourier transform [arXiv:2505.12615]

QSP reference: [arXiv:2510.00443]

Section 4

Application: Hamiltonian simulation

Hamiltonian simulation

$$\frac{d}{dt}u(t) = -iHu(t), \quad 0 \leq t \leq T, \quad H \text{ Hermitian}$$

- Unitary evolution: e^{-iHt}
- A natural and promising application of quantum computers
- Algorithms: product formula, quantum walk, truncated Taylor series, QSVT, randomization, ...

QSVT for Hamiltonian simulation

$U(T) = e^{-iHT}$. Consider $f(x) = e^{-ix\|H\|T} = \cos(\|H\|Tx) - i\sin(\|H\|Tx)$.
Polynomial approximations:

- Taylor: $\cos(\|H\|Tx) \approx \sum_{k=0}^{O(\|H\|T+\log(1/\epsilon))} \frac{(-1)^k \|H\|^{2k} T^{2k}}{(2k)!} x^{2k}$
- Jacobi-Anger: $\cos(Tx) \approx J_0(\|H\|T) + 2\sum_{k=1}^{O(\|H\|T+\log(1/\epsilon))} (-1)^k J_{2k}(\|H\|T)\, T_{2k}(x)$

QSVT for Hamiltonian simulation

Algorithm: use QSVT to implement $\cos(TH)$ and $\sin(TH)$, then LCU to combine.
Query complexity:

$$O\left(\|H\| T + \log \tfrac{1}{\epsilon}\right)$$

- Optimal scaling
- Drawback: cannot exploit structure inside $H = \sum_\gamma H_\gamma$, $H = -\Delta_x + V(x)$

Product formula

$H = A + B$. Split evolution:

$$e^{-iHT} = \lim_{L\to\infty} \left(e^{-iBT/L} e^{-iAT/L}\right)^L$$

- 1st order: $U_1(t) = e^{-iBt}e^{-iAt} \approx e^{-iHt} + O(t^2)$
- 2nd order: $U_2(t) = e^{-iAt/2}e^{-iBt}e^{-iAt/2} \approx e^{-iHt} + O(t^3)$
- Suzuki recursion: $U_{2p+2}(t) = U_{2p}(s_p t)^2 U_{2p}((1-4s_p)t) U_{2p}(s_p t)^2$

Complexity (commutator scaling, suboptimal in T, ϵ):

$$O\left(\frac{\alpha_{\text{comm}}^{1/p} T^{1+1/p}}{\epsilon^{1/p}}\right)$$

Multi-product formula

Simultaneously achieve commutator scaling and optimal time/precision via extrapolation:

$$U_{\mathrm{MPF}}(t) = \sum_{j=1}^{M} a_j U_2(t/k_j)^{k_j}$$

Time-dependent Hamiltonian simulation

$$\frac{d}{dt}u(t) = -iH(t)u(t), \quad 0 \le t \le T, \quad H(t) \text{ Hermitian}$$

- Product formula: commutator scaling (including derivatives), suboptimal in T, ϵ
- Truncated Dyson series: near-optimal

$$O\left(\max_t \|H(t)\| T \log \frac{\max_t \|H(t)\| T}{\epsilon}\right)$$

Reference: [arXiv:2411.03180]

Section 5

Application: Linear systems of equations

Quantum linear system problem (QLSP)

Classical: given $N \times N$ Hermitian A and vector b, compute $x = A^{-1}b$.

Non-Hermitian: embed as $\begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix} \begin{pmatrix} 0 \\ x \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}$.

Quantum: find ϵ-approximation of $|x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|}$.

Key parameters: error ϵ, condition number $\kappa = \|A\|\|A^{-1}\|$.

QSVT for QLSP

$A = W\Sigma V^\dagger \to A^{-1} = V\Sigma^{-1}W^\dagger$. Consider $f(x) = \frac{1}{\kappa x}$.
Polynomial approx: odd polynomial $p(x)$ of degree $O(\kappa \log(1/\epsilon))$ with $|p| \leq 1$ and $|p(x) - 1/(\kappa x)| \leq \epsilon$ for $x \in [1/\kappa, 1]$.
Algorithm: QSVT to implement $p(A^\dagger)$, apply to $|0\rangle |b\rangle$, post-select.

Harrow-Hassidim-Lloyd (HHL)

The first quantum algorithm for QLSP.[7]
Let $(\lambda_j, |v_j\rangle)$ be eigenpairs of A and $|b\rangle = \sum_j \beta_j |v_j\rangle$:

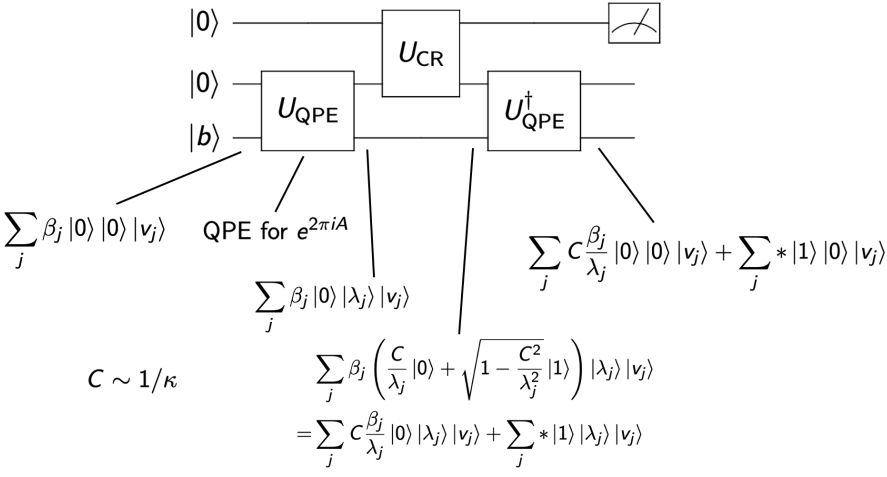
$$A^{-1}|b\rangle = \left(\sum_j \lambda_j^{-1} |v_j\rangle \langle v_j|\right)\left(\sum_j \beta_j |v_j\rangle\right) = \sum_j \frac{\beta_j}{\lambda_j}|v_j\rangle$$

Subroutines:

- **QPE:** $U_{\mathrm{QPE}}|\psi\rangle|0\rangle = |\psi\rangle|\theta\rangle$ where $U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$
- **Controlled rotation:** $U_{\mathrm{CR}}|\theta\rangle|0\rangle = |\theta\rangle\left(f(\theta)|0\rangle + \cdots\right)$

[7] [arXiv:0811.3171]

HHL


$|0\rangle$
$|0\rangle$
$|b\rangle$
U_{QPE}
U_{CR}
$U_{QPE}^{\dagger}$
$\sum_j \beta_j |0\rangle |0\rangle |v_j\rangle$
QPE for $e^{2\pi i A}$
$\sum_j \beta_j |0\rangle |\lambda_j\rangle |v_j\rangle$
$\sum_j C\frac{\beta_j}{\lambda_j} |0\rangle |0\rangle |v_j\rangle + \sum_j * |1\rangle |0\rangle |v_j\rangle$
$C \sim 1/\kappa$
$\sum_j \beta_j \left(\frac{C}{\lambda_j} |0\rangle + \sqrt{1 - \frac{C^2}{\lambda_j^2}} |1\rangle\right) |\lambda_j\rangle |v_j\rangle$
$= \sum_j C\frac{\beta_j}{\lambda_j} |0\rangle |\lambda_j\rangle |v_j\rangle + \sum_j * |1\rangle |\lambda_j\rangle |v_j\rangle$

LCU: Fourier approach

Key identity: $\frac{1}{x} = \frac{i}{\sqrt{2\pi}} \int_0^\infty dy \int_{-\infty}^\infty dz\, z e^{-z^2/2} e^{-ixyz}$.

For Hermitian A:

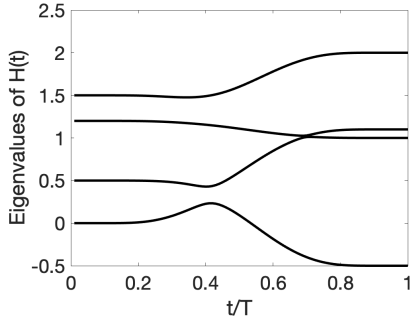
$$A^{-1} \approx \sum_{j,j'} c_{j,j'} e^{-iy_j z_{j'} A}$$

implemented by LCU + Hamiltonian simulation.

Adiabatic Quantum Computing (AQC)

$$i\partial_t |\psi(t)\rangle = H(t/T)\,|\psi(t)\rangle\,, \quad t \in [0, T]$$

- Eigenvector of $H(0) \rightarrow$ eigenvector of $H(1)$ if T large enough and gap condition holds
- Application: quantum computing model for eigenvalue problems



AQC for QLSP

$$H_0 = \begin{pmatrix} 0 & Q_b \\ Q_b & 0 \end{pmatrix}, \quad H_1 = \begin{pmatrix} 0 & AQ_b \\ Q_bA & 0 \end{pmatrix}, \quad Q_b = I - |b\rangle \langle b|$$

$$H(s) = (1 - f(s))H_0 + f(s)H_1$$

- Eigenpath at eigenvalue 0 connects $[b; 0]$ and $[x; 0]$
- Need to carefully choose scheduling function $f(s)$

Complexities

Method	Circuit (A)	Circuit (b)
HHL [0811.3171]	$O(\kappa^2/\epsilon)$	$O(\kappa)$
LCU [1511.02306]	$O(\kappa^2 \mathrm{polylog}(\kappa/\epsilon))$	$O(\kappa)$
QSVT [1804.01973]	$O(\kappa^2 \log(\kappa/\epsilon))$	$O(\kappa)$
RM [1805.10549]	$O((\kappa/\epsilon)\log\kappa)$	$O((\kappa/\epsilon)\log\kappa)$
AQC(p) [1909.05500]	$O((\kappa/\epsilon)\log(\kappa/\epsilon))$	$O((\kappa/\epsilon)\log(\kappa/\epsilon))$
AQC(exp) [1909.05500]	$O(\kappa\,\mathrm{polylog}(\kappa/\epsilon))$	$O(\kappa\,\mathrm{polylog}(\kappa/\epsilon))$
dAQC+EF [2111.08152]	$O(\kappa\log(1/\epsilon))$	$O(\kappa\log(1/\epsilon))$
KR [2406.12086]	$O(\kappa\log(1/\epsilon))$	$O(\kappa\log(1/\epsilon))$
Tunable VTAA [2410.18178]	$O(\kappa\log(\kappa/\epsilon)\log(1/\epsilon))$	$O(\kappa)$

Section 6

Summary

Summary

Quantum computing: use quantum mechanics to perform computing.
Input models: Vector: quantum state Matrix: block-encoding
Algorithms:

- LCU
- Matrix functions: QSVT can implement degree-d polynomials with cost $O(d)$
- Hamiltonian simulation: QSVT, product formula
- Linear systems: QSVT, HHL, LCU, adiabatic

Additional resources

- IPAM Tutorial: Mathematical and Computational Challenges in Quantum Computing (Sept. 12–15, 2023)
  https://www.ipam.ucla.edu/programs/workshops/mathematical-and-computational-challenges-in-quantum-computing-tutorials/
- Lecture notes by Lin Lin:
  https://math.berkeley.edu/~linlin/qasc/

Thank you!

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Matrix addition

Goal: block-encode $\sum_j c_j A_j$ for $\|A_j\| \leq 1$.
Algorithm: LCU with U_{A_j} as block-encodings.

$$O_s = \sum_j |j\rangle \langle j| \otimes U_{A_j}, \qquad U_{A_j} = \begin{pmatrix} A_j & * \\ * & \end{pmatrix}$$

Vector addition

Goal: compute $\sum_j c_j |u_j\rangle$.
Algorithm: LCU with state prep oracles $U_j : |0\rangle \to |u_j\rangle$:

$$\sum_j c_j U_j |0\rangle = \sum_j c_j |u_j\rangle$$

Matrix multiplication

Two block-encodings U_{A_0}, U_{A_1}. Naïve stacking does *not* work:

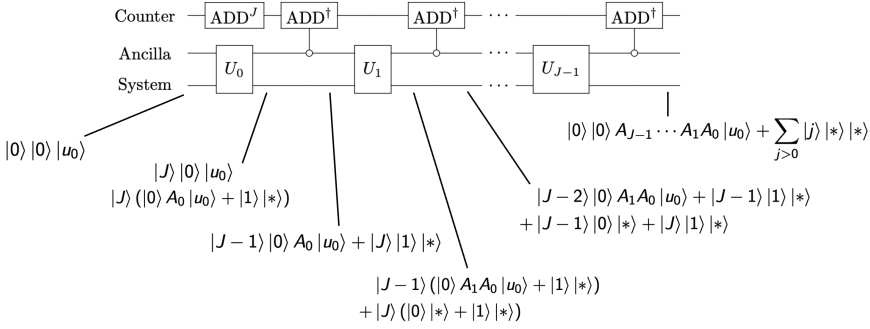
$$U_{A_1} U_{A_0} |0\rangle |u_0\rangle \neq |0\rangle A_1 A_0 |u_0\rangle + \cdots$$

Method 1: duplicate ancilla qubits. Multiplication of J matrices uses $O(J)$ extra ancilla qubits.

Matrix multiplication

Method 2: compression gadget

(Low-Wiebe [arXiv:1805.00675]; Fang-Lin-Tong [arXiv:2208.06941])



$$\text{ADD} : |0\rangle \to |1\rangle \to |2\rangle \to \cdots \to |J\rangle \to |0\rangle$$

Qubitization (non-Hermitian block-encoding)

For general (non-Hermitian) U_A:

$$U_A |0\rangle |v_j\rangle = \lambda_j |0\rangle |v_j\rangle + \sqrt{1-\lambda_j^2}\, |\perp_j'\rangle\,, \qquad U_A^\dagger |0\rangle |v_j\rangle = \lambda_j |0\rangle |v_j\rangle + \sqrt{1-\lambda_j^2}\, |\perp_j\rangle$$

$$[U_A]_{\mathcal{H}_j \to \mathcal{H}_j'} = [U_A^\dagger]_{\mathcal{H}_j' \to \mathcal{H}_j} = \begin{pmatrix} \lambda_j & \sqrt{1-\lambda_j^2} \\ \sqrt{1-\lambda_j^2} & -\lambda_j \end{pmatrix}$$

$$(U_A^\dagger Z_\Pi U_A Z_\Pi)^k = \begin{pmatrix} T_{2k}(A) & * \\ & * \end{pmatrix}$$

Quantum signal processing (QSP)

$$U_\Phi(\lambda) = e^{i\phi_0 Z} \prod_{j=1}^{d} \left[U(\lambda) e^{i\phi_j Z} \right] = \begin{pmatrix} P(\lambda) & * \\ & * \end{pmatrix}$$

Using $Ze^{i\phi Z} = (-i)e^{i(\phi+\pi/2)Z}$, we can absorb Z into modified phases:

- Even d: $\tilde{\phi}_j = \phi_j + (-1)^j \pi/2$ for $j \neq 0$; $\tilde{\phi}_0 = \phi_0$
- Odd d: same, with an extra Z that doesn't affect the top-left element

QSP + Qubitization

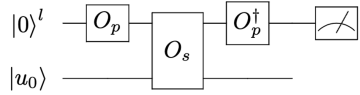
For Hermitian A and polynomial $P(x)$ satisfying QSP conditions:

- Even d: $e^{i\tilde{\phi}_0 Z_\Pi} \prod_{j=1}^{d/2} \left[U_A^\dagger e^{i\tilde{\phi}_{2j-1} Z_\Pi} U_A e^{i\tilde{\phi}_{2j} Z_\Pi} \right] = \begin{pmatrix} P(A) & * \\ * & \end{pmatrix}$
- Odd d: similar with alternating U_A, $U_A^\dagger$

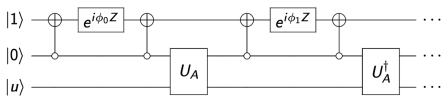
QSP/QSVT vs LCU

- Both can implement matrix functions
- LCU has overhead from ℓ^1-norm of coefficients and extra control logic
- For general matrices:
  - QSVT: singular value transformation
  - LCU: eigenvalue transformation via $f(A) = \frac{1}{2\pi i} \oint_\Gamma f(z)(z - A)^{-1} dz$

LCU:



QSP:



HHL: complexity

Two sources:

- Cost per run (mainly QPE): $O(\kappa/\epsilon)$
- Number of repeats: $O(1/(C\|A^{-1}\left|b\right\rangle\|)) = O(\kappa)$
- Overall: $O(\kappa^2/\epsilon)$

Subroutines needed: Hamiltonian simulation ($e^{2\pi iA}$) and QPE.

LCU: Chebyshev approach

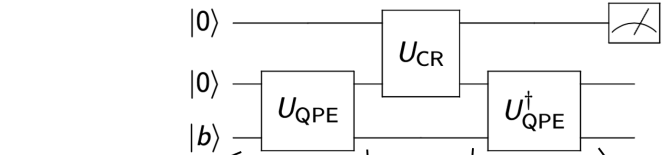
Expand $1/x$ in Chebyshev polynomials:

$$A^{-1} \approx \sum_{j=0}^{O(\kappa \operatorname{polylog}(\kappa/\epsilon))} c_j T_{2j+1}(A)$$

$\|T_{2j+1}(A)\| \leq 1$ but not unitary $\Rightarrow$ need block-encoding.
Same overall complexity:
$\kappa \operatorname{polylog}(\kappa/\epsilon) \times \kappa/\sqrt{\log(\kappa/\epsilon)} = O(\kappa^2 \operatorname{polylog}(\kappa/\epsilon))$.

$$\begin{aligned}
\frac{1}{x} &\approx \frac{1-(1-x^2)^d}{x} && (d \sim \kappa^2 \log(\kappa/\epsilon)) \\
&= 4\sum_{j=0}^{d-1}(-1)^j \left(2^{-2d}\sum_{i=j+1}^{d}\binom{2d}{d+i}\right) T_{2j+1}(x) \\
&\approx 4\sum_{j=0}^{J}(-1)^j \left(2^{-2d}\sum_{i=j+1}^{d}\binom{2d}{d+i}\right) T_{2j+1}(x) && (J \sim \sqrt{d\log(d/\epsilon)})
\end{aligned}$$

HHL circuit



$\sum_j \beta_j |0\rangle |0\rangle |v_j\rangle$

QPE for $e^{2\pi i A}$

$\sum_j \beta_j |0\rangle |\lambda_j\rangle |v_j\rangle$

$\sum_j C\frac{\beta_j}{\lambda_j} |0\rangle |0\rangle |v_j\rangle + \sum_j * |1\rangle |0\rangle |v_j\rangle$

$C \sim 1/\kappa$

$$\sum_j \beta_j \left(\frac{C}{\lambda_j} |0\rangle + \sqrt{1 - \frac{C^2}{\lambda_j^2}} |1\rangle \right) |\lambda_j\rangle |v_j\rangle$$

$$= \sum_j C\frac{\beta_j}{\lambda_j} |0\rangle |\lambda_j\rangle |v_j\rangle + \sum_j * |1\rangle |\lambda_j\rangle |v_j\rangle$$

Quantum Adiabatic Theorem

Theorem (Jansen-Ruskai-Seiler)

With gap $\Delta(s)$:

$$\eta(s) \leq C\left\{\frac{\|H'(0)\|^2}{T\Delta^2(0)} + \frac{\|H'(s)\|^2}{T\Delta^2(s)} + \frac{1}{T}\int_0^s \left(\frac{\|H''(\tau)\|^2}{\Delta^2(\tau)} + \frac{\|H'(\tau)\|^4}{\Delta^3(\tau)}\right) d\tau\right\}$$

- Error $\leq \epsilon$: $T = O(\Delta_*^{-3}\epsilon^{-1})$ — cubic dependence on gap
- In QLSP: $\Delta_* \sim 1/\kappa \Rightarrow T = O(\kappa^3\epsilon^{-1})$

Time-optimal AQC

Interpolate $H(s) = (1 - f(s))H_0 + f(s)H_1$; choose $f(s)$ to slow down when gap is small.
AQC(p): $\dot{f}(s) = c\,\Delta^p(f(s)) \Rightarrow T = O(\kappa/\epsilon)$.[8]

[8] An-Lin [arXiv:1909.05500]

AQC(exp)

Improved adiabatic theorem: error $\sim O(T^{-k})$ with boundary cancellation.
AQC(exp): $f(s) = c^{-1} \int_0^s \exp[-u^{-1}(1-u)^{-1}]du \Rightarrow T = O(\kappa \,\mathrm{polylog}(\kappa/\epsilon))$.[9]

[9] An-Lin [arXiv:1909.05500]

Preconditioning

$\mathcal{A}x = b \Leftrightarrow M\mathcal{A}x = Mb$; effective if $\kappa(M\mathcal{A}) \ll \kappa(\mathcal{A})$.
Classical: diagonal, incomplete factorization, SPAI, ...
Quantum: SPAI [arXiv:1301.2340], circulant [Shao-Xiang].

Preconditioning: $(A + B)\,|x\rangle \approx |b\rangle$

Assume A easily invertible, large $\|A\|$, moderate $\|B\|, \|A^{-1}\|, \|(A+B)^{-1}\|$.
$\kappa(A+B) \sim O(\|A\|)$. Preconditioner A^{-1}: $\kappa(I + A^{-1}B) = O(1)$.
Algorithm pipeline:

$$A^{-1} \to A^{-1}B \to I + A^{-1}B \to (I + A^{-1}B)^{-1} \to (I + A^{-1}B)^{-1}A^{-1} = (A+B)^{-1}$$

Needs: matrix addition, multiplication, and fast inversion of a diagonal matrix.